UNIVERSITY OF BRISTOL

Winter 2025 Examination Period

SCHOOL OF COMPUTER SCIENCE

Third Year Examination for the Degrees
of
Bachelor of Science
Master of Engineering
Master of Science

Types and Lambda Calculus

TIME ALLOWED:
1 Hour

Answers to COMS30040-W: Types and Lambda Calculus

Intended Learning Outcomes:

Q1. Credit will be given for partially correct answers. You may use any result from the course material, as long as it is labelled clearly. A reminder of key definitions is provided at the back of this section.

* (a) For each of the following terms, either give its normal form, or state that it is not normalisable.

i. $(\lambda xy.\, x)\ \underline{2}\ \underline{3}$

ii. $\mathsf{ifz}\ ((\lambda x.\, \mathsf{S}\ (\mathsf{S}\ x))\ \mathsf{Z})\ \underline{2}\ \underline{3}$

iii. $(\lambda xy.\, x\ (y\ (y\ \underline{2})))\ (\lambda z.\, z)\ \mathsf{S}$

iv. $\mathsf{fix}\ (\lambda xy.\, x)\ x_1\ x_2\ x_3\ x_4$

v. $\mathsf{fix}\ (\lambda yx.\, x)\ x_1\ x_2\ x_3\ x_4$

[10 marks]

Solution:

i. $\underline{2}$

ii. $\underline{3}$

iii. $\underline{4}$

iv. not normalisable

v. $x_1\ x_2\ x_3\ x_4$

* (b) For each of the following judgements, give a type derivation:

i. $x : (a \to a) \to b \to c \vdash \lambda y.\, x\ (\lambda z.\, z)\ y : b \to c$

ii. $\vdash \lambda f x.\, \mathsf{ifz}\ x\ x\ (f\ x) : (\mathsf{Nat} \to \mathsf{Nat}) \to \mathsf{Nat} \to \mathsf{Nat}$

iii. $\vdash \mathsf{fix}\ (\lambda yx.\, x) : \mathsf{Nat} \to \mathsf{Nat}$

[12 marks]

Solution:

i. With $\Gamma = \{\, x : (a \to a) \to b \to c,\ y : b \,\}$

$$\dfrac{\dfrac{\dfrac{}{\Gamma \vdash x : (a \to a) \to b \to c} \quad \dfrac{\dfrac{}{\Gamma,\ z : a \vdash z : a}}{\Gamma \vdash \lambda z.\, z : a \to a}}{\Gamma \vdash x\ (\lambda z.\, z) : b \to c} \quad \dfrac{}{\Gamma \vdash y : b}}{\dfrac{\Gamma \vdash x\ (\lambda z.\, z)\ y : c}{x : (a \to a) \to b \to c \vdash \lambda y.\, x\ (\lambda x.\, x)\ y : b \to c}}$$

ii. With $\Gamma = \{ f : \mathsf{Nat} \to \mathsf{Nat},\ x : \mathsf{Nat} \}$

$$\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{}{\Gamma \vdash \mathsf{ifz} : \mathsf{Nat} \to \mathsf{Nat} \to \mathsf{Nat} \to \mathsf{Nat}} \quad \dfrac{}{\Gamma \vdash x : \mathsf{Nat}}}{\Gamma \vdash \mathsf{ifz}\ x : \mathsf{Nat} \to \mathsf{Nat} \to \mathsf{Nat}} \quad \dfrac{}{\Gamma \vdash x : \mathsf{Nat}}}{\Gamma \vdash \mathsf{ifz}\ x\ x : \mathsf{Nat} \to \mathsf{Nat}}}{\Gamma \vdash \mathsf{ifz}\ x\ x\ (f\ x) : \mathsf{Nat}}}{f : \mathsf{Nat} \to \mathsf{Nat} \vdash \lambda x.\, \mathsf{ifz}\ x\ x\ (f\ x) : \mathsf{Nat} \to \mathsf{Nat}}}{\vdash \lambda f x.\, \mathsf{ifz}\ x\ x\ (f\ x) : (\mathsf{Nat} \to \mathsf{Nat}) \to \mathsf{Nat} \to \mathsf{Nat}}$$

D

where D is:

$$\dfrac{\dfrac{}{\Gamma \vdash f : \mathsf{Nat} \to \mathsf{Nat}} \quad \dfrac{}{\Gamma \vdash x : \mathsf{Nat}}}{\Gamma \vdash f\ x : \mathsf{Nat}}$$

iii. With $A = ((\mathsf{Nat} \to \mathsf{Nat}) \to \mathsf{Nat} \to \mathsf{Nat}) \to \mathsf{Nat} \to \mathsf{Nat}$

$$\dfrac{\dfrac{\dfrac{\dfrac{}{y : \mathsf{Nat} \to \mathsf{Nat},\ x : \mathsf{Nat} \vdash x : \mathsf{Nat}}}{y : \mathsf{Nat} \to \mathsf{Nat} \vdash \lambda x.\, x : \mathsf{Nat} \to \mathsf{Nat}}}{\lambda y x.\, x : (\mathsf{Nat} \to \mathsf{Nat}) \to \mathsf{Nat} \to \mathsf{Nat}} \quad \vdash \mathsf{fix} : A}{\vdash \mathsf{fix}\ (\lambda y x.\, x) : \mathsf{Nat} \to \mathsf{Nat}}$$

**(c) Prove that the following property of terms is undecidable:

$$\{ M \in \Lambda \mid M \approx M\ M \}$$

[6 marks]

Solution: The result follows from the Scott-Curry Theorem. Let us call the set X

- To see the set is nontrivial, observe that $\underline{3} \notin X$ and $\underline{\mathsf{id}} \in X$.
- To see the set is closed under convertibility, suppose $M \in X$ and $M \approx N$. From the former, $M \approx M\ M$ and hence $N \approx M\ M$. Since $M\ M \approx N\ N$, $N \approx N\ N$ and so $N \in X$, as required.

**(d) Prove that the term $\lambda x.\, x\ (\underline{\mathsf{id}}\ x)$ is untypable.

[7 marks]

Solution: We suppose $\lambda x.\, x\ (\underline{\mathsf{id}}\ x)$ is typable and try to obtain a contradiction. Then there is a type A such that $\vdash x\ (\underline{\mathsf{id}}\ x) : A$. By the Preservation theorem, therefore $\vdash x\ x : A$, but $x\ x$ is known to be untypable. (This proof can also be done, without reference to Preservation, just using inversion.)

*** (e) Let us say that a closed term M is *solvable* just if, for all terms P, there exists a (possibly empty) sequence of terms $N_1, \ldots, N_n$ such that: $M\ N_1\ \cdots N_n\ \approx\ P$.

i. Prove that the term $\lambda xy.\, y\, x\, x$ is solvable.

ii. Give an example of a term that is solvable but does not have a normal form.

iii. Is the following set decidable? Justify your answer.

$$\{\, M \in \Lambda \mid M \text{ is closed, } \textit{pure} \text{ and (solvable or not normalisable)} \,\}$$

[10 marks]

Solution:

i. Let P be a term. Then let the witness be $N_1 = \underline{\mathsf{div}}$, $N_2 = (\lambda z_1 z_2.\, P)$, so that $(\lambda xy.\, y\ x\ x)\ N_1\ N_2\ \rhd^*\ (\lambda z_1 z_2.\, P)\ N_1\ N_1\ \rhd^*\ P$.

ii. $\lambda xy.\, y\ \underline{\mathsf{div}}$

iii. The set is decidable because it is exactly all closed pure terms, and closed pure terms can be recognised by parsing. To see this we show that every *closed*, *pure* term in normal form is solvable, and therefore, every pure term that has a normal form is solvable. So let M be closed and pure and let P be a term. Then M has shape $\lambda x_1 \ldots x_n.\, y\ M_1\ \cdots M_k$ for $n \geq 0$ and $k \geq 0$. Since M is closed, y is one of the parameters x_i, say x_j. Then we have:

$$(\lambda x_1 \ldots x_n.\, x_j\ M_1\ \cdots M_k)\ y_1\ \cdots\ y_{j-1}\ (\lambda z_1 \ldots z_k.\, P)\ y_{j+1}\ \cdots y_n$$

which reduces to P, as required.

*** (f) Let $\#$ be a Gödel numbering of PCF terms. Without using the Second Recursion Theorem, prove that there is a term M satisfying $M\ \approx\ \underline{\#M}$

[5 marks]

Solution: To see this, first define N as $\lambda x.\, \underline{\mathsf{app}}\ x\ (\underline{\mathsf{gnum}}\ x)$. Then set M to be $N\ \underline{\#N}$. It follows that:

$$\begin{array}{rcl} N\ \underline{\#N} & \approx & \underline{\mathsf{app}}\ \underline{\#N}\ (\underline{\mathsf{gnum}}\ \underline{\#N}) \\ & \approx & \underline{\mathsf{app}}\ \underline{\#N}\ (\underline{\#\#N}) \\ & \approx & \underline{\#(N\ \underline{\#N})} \end{array}$$

Reference for Types and Lambda Calculus

Terms

$$\begin{array}{rrcl} \text{(Terms)} & M,\, N & ::= & x \mid c \mid (\lambda x.\, M) \mid (MN) \\ \text{(Constants)} & c & ::= & \mathsf{fix} \mid \mathsf{Z} \mid \mathsf{S} \mid \mathsf{pred} \mid \mathsf{ifz} \end{array}$$

The set of all terms is written Λ. A term is said to be *pure* just if it contains no constants. Abbreviations:

$$\begin{array}{rcl} \underline{n} & = & \mathsf{S}^n\ \mathsf{Z} \\ \underline{\mathsf{id}} & = & \lambda x.\, x \\ \underline{\mathsf{const}} & = & \lambda xy.\, x \\ \underline{\mathsf{sub}} & = & \lambda xyz.\, xz(yz) \\ \underline{\mathsf{div}} & = & \mathsf{fix}\ \underline{\mathsf{id}} \end{array}$$

Free Variables

$$\begin{array}{rcl} \mathsf{FV}(x) & = & \{x\} \\ \mathsf{FV}(c) & = & \emptyset \\ \mathsf{FV}(PQ) & = & \mathsf{FV}(P) \cup \mathsf{FV}(Q) \\ \mathsf{FV}(\lambda x.\, N) & = & \mathsf{FV}(N) \setminus \{x\} \end{array}$$

A term M is said to be *closed* just if $\mathsf{FV}(M) = \emptyset$.

Substitution

$$\begin{array}{rcll} c[N/x] & = & c & \\ y[N/x] & = & y & \text{if } x \neq y \\ y[N/x] & = & N & \text{if } x = y \\ (PQ)[N/x] & = & P[N/x]Q[N/x] & \\ (\lambda y.\, P)[N/x] & = & \lambda y.\, P & \text{if } y = x \\ (\lambda y.\, P)[N/x] & = & \lambda y.\, P[N/x] & \text{if } y \neq x \text{ and } y \notin \mathsf{FV}(N) \end{array}$$

Redexes

$$\begin{array}{r} \mathsf{pred}\ \mathsf{Z} \ /\ \mathsf{Z} \\ \mathsf{pred}\ (\mathsf{S}\ N) \ /\ N \\ \mathsf{ifz}\ \mathsf{Z}\ N\ P \ /\ N \\ \mathsf{ifz}\ (\mathsf{S}\ M)\ N\ P \ /\ P \\ (\lambda x.\, M)\ N \ /\ M[N/x] \\ \mathsf{fix}\ M \ /\ M\ (\mathsf{fix}\ M) \end{array}$$

One Step

$$C[] ::= [] \mid M\ C[] \mid C[]\ N \mid \lambda x.\, C[]$$

Define $M \rhd N$ just if there is a context $C[]$ and a redex/contraction pair P / Q such that $M = C[P]$ and $N = C[Q]$.

Reduction and Conversion

- $P \rhd^0 Q$ just if $P = Q$.
- $P \rhd^{k+1} Q$ just if there is some U such that $P \rhd^k U$ and $U \rhd Q$.

Define $M \rhd^* N$ just if there is some n such that $M \rhd^n N$.

We write $M \approx N$ just if there is a term P such that $M \rhd^* P$ and $N \rhd^* P$.

- If $M \rhd^* N$ then the term N is said to be a **reduct** of M.
- If $M \rhd^* N$ and $M \neq N$ then the term N is said to be a **proper reduct** of M.
- A term M without a proper reduct is a **normal form**.
- A term M that can reduce to normal form **has a normal form** or is **normalisable**.
- A term M that has no infinite reduction sequences is said to be **strongly normalisable**.

Computability

A **Gödel numbering** is a pair of computable (partial) functions:

- $\# : \Lambda \to \mathbb{N}$
- $\#^{-1} : \mathbb{N} \to \Lambda$

with the property that $\#^{-1}\ (\#\ M) = M$.

Let $\Phi \subseteq \Lambda$ be a property of PCF-terms. We say that Φ is **decidable** just if the characteristic function of Φ is computable, i.e. there is an algorithm for computing $\chi : \Lambda \to \mathbb{N}$ satisfying, for all terms $M \in \Lambda$:

$$\chi(M) = \begin{cases} 1 & \text{if } M \in \Phi \\ 0 & \text{if } M \notin \Phi \end{cases}$$

Type Assignment

$$(\text{Types}) \quad A,\, B \quad ::= \mathsf{Nat} \mid a \mid (A \to B)$$

Let $\mathbb{C}$ be the following collection of type assignments:

$$\begin{aligned}\{\mathsf{Z} : \mathsf{Nat}\} \cup \{\mathsf{S} : \mathsf{Nat} \to \mathsf{Nat}\} \cup \{\mathsf{pred} : \mathsf{Nat} \to \mathsf{Nat}\} \\ \cup \{\mathsf{ifz} : \mathsf{Nat} \to A \to A \to A \mid A \in \mathbb{T}\} \\ \cup \{\mathsf{fix} : (A \to A) \to A \mid A \in \mathbb{T}\}\end{aligned}$$

The typing rules are:

$$x{:}A \in \Gamma \frac{}{\Gamma \vdash x : A}\,(\text{TVar}) \qquad c{:}A \in \mathbb{C} \frac{}{\Gamma \vdash c : A}\,(\text{TCst})$$

$$\frac{\Gamma \vdash M : B \to A \quad \Gamma \vdash N : B}{\Gamma \vdash MN : A}\,(\text{TApp}) \qquad x \notin \mathsf{dom}\ \Gamma \frac{\Gamma \cup \{x : B\} \vdash M : A}{\Gamma \vdash \lambda x.\, M : B \to A}\,(\text{TAbs})$$

We say that a closed term M is **typable** just if there is some type A such that $\vdash M : A$ is derivable in the type system. If $\vdash M : A$, then M is said to be an **inhabitant** of A. The **pure-term inhabitation problem**, is the problem of, given a type A, determining if there a closed, *pure* term M such that $\vdash M : A$.

Useful Theorems

Theorem (Confluence)**.** *If $M \rhd^* P$ and $M \rhd^* Q$ then there exists a term N such that $P \rhd^* N$ and $Q \rhd^* N$.*

Theorem (Unique Normal Forms)**.** *If $M \rhd^* N$ and $M \rhd^* P$ with N and P normal forms, then $N = P$.*

Theorem (Scott-Curry)**.** *Let $\Phi \subseteq \Lambda$ satisfy both the following properties:*

- $\emptyset \neq \Phi \neq \Lambda$
- *if $M \in \Phi$ and $M \approx N$, then $N \in \Phi$*

Then it follows that Φ is undecidable.

Theorem (Preservation)**.** *If $\Gamma \vdash M : A$ and $M \rhd N$ then $\Gamma \vdash N : A$*

Theorem (Progress)**.** *If $\vdash M : A$ then either M is a value or there is some N such that $M \rhd N$.*